

On the minimum density of 2-clique and 3/2-clique in directed graphs

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Abstract

For an integer k , a k -clique is a graph in which the distance from one node to any other is at most k . The concept has been used to find cohesive groups in social network analysis. Traditionally, the k -clique is defined on undirected graphs, i.e., the relation is reciprocal. In this paper, we study the minimum density of a 2-clique when the graph is directed. In a directed graph, it is also possible to define a 3/2-clique, which is a middle ground between 1-clique and 2-clique. We show that the minimum densities of a 3/2-clique and a 2-clique with size n is respectively 0.5 and $2/n$. The result ensures the worst case cohesion of the groups found by means of k -cliques in social networks with unidirectional relationship.

Keywords: social network analysis, graph theory, clique, transitive extension

1 Introduction

A social network is a social structure between actors (people, organizations or other social entities). The nodes in the network are the actors, while the links show some kind of social ties, such as friendship, co-working or message flow. One of the interesting problems in social network analysis is how to find cohesive groups in a given network. This problem can be regarded as a restricted version of a more generalized form called as *community detection*, which is also known as *graph clustering* [5].

In graph theory, a clique is a complete graph/subgraph, i.e., each node is connected to all other nodes in the clique. Since a clique is the most cohesive group of nodes, a straightforward method looks for cohesive groups by means of cliques. However, it usually doesn't work in both real-life and online data, not only due to the

hardness in computation (Maximum Clique is a well-known NP-hard problem [1]) but also that the real data is never perfect.

In the literature, one of the methods to find groups is by means of 2-clique or even k -clique for some integer $k > 2$, which are relaxations of clique [2, 3, 4]. A substructure of a graph is a k -clique if the distance between any pair of nodes is at most k . Traditionally, the k -clique is defined on undirected graphs, i.e., the relationship is reciprocal (symmetric). In fact, many of the previous researches in social network analysis focus on reciprocal relation ties. However, there are directed social networks, in which the relationship is unidirectional. For example, the friend network in blogosphere is usually directed. In most of the blog (Web-log) systems, the friend lists are maintained by the bloggers themselves, i.e., a blogger may add any other as a friend without the other's confirmation.

For the undirected case, it is easy to see that the minimum number of edges for an n -node 2-clique is $n - 1$. The reason is that we need at least $n - 1$ edges to keep the graph connected. This bound is also tight since a *star* is a 2-clique and has exactly $n - 1$ edges. In this paper, we study the minimum density of a 2-clique when the graph is directed.

Another interesting thing is that, in a directed graph, it is also possible to define a 3/2-clique, which is a middle ground between 1-clique and 2-clique. A directed graph is a 3/2-clique if $d(u, v) + d(v, u) \leq 3$ for any nodes u and v , in which $d(u, v)$ denote the distance from u to v . We can see that it is meaningless for the undirected case.

For a directed graph with n nodes and m edges, its density is defined by $\frac{m}{n(n-1)}$, which is the ratio of the number of edges to the number of all possible edges. Interestingly we show that the minimum densities of 3/2-clique and 2-clique are respectively $1/2$ and $2/n$. The result ensures the worst case cohesion of the groups found by means of k -cliques in social networks with unidirectional relationship. The newly defined 3/2-clique also

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provides an alternative that can be used to find cohesive groups in directed social networks, and the constant lower bound of its density ensures its worst case quality.

2 The density analysis

In the following paragraphs, a graph is a directed simple graph and is assumed to be connected. The density of a graph/subgraph is the ratio of the number of its arcs to the number of all possible arcs. In this paper, self-loops, something like (u, u) , are not allowed. For a directed graph with n nodes, the number of all possible arcs is $n(n-1)$. We first consider the $3/2$ -clique. Let $\sigma_k(n)$ denote the minimum number of arcs of any n -node k -clique and $\delta_k(n) = \sigma_k(n)/(n(n-1))$ the minimum density. In this paper, we assume that $n \geq 3$.

Lemma 1: $\delta_{3/2}(n) \geq 1/2$ for any $n \geq 3$.

Proof: For any pair of nodes u and v , if neither (u, v) nor (v, u) exists, we have that $d(u, v) > 1$ and $d(v, u) > 1$, and consequently $d(u, v) + d(v, u) > 3$, i.e., they cannot be in the same $3/2$ -clique. Therefore the number of arcs in a $3/2$ -clique is at least $n(n-1)/2$. $\square$

Lemma 2: $\sigma_{3/2}(n+2) \leq \sigma_{3/2}(n) + 2n + 1$ for any $n \geq 2$.

Proof: Suppose that G is a $3/2$ -clique of size n . We construct a $3/2$ -clique of size $n+2$ by adding $2n+1$ arcs as follows. Let u and v be two nodes not in G . For each node x in G , we add arcs (x, u) and (v, x) . There is another arc (u, v) . We claim that the resulting graph is a $3/2$ -clique since (x, u, v) is a 3-cycle for each node x in G . $\square$

Theorem 3: For $n \geq 3$ and $n \neq 4$, $\delta_{3/2}(n) = 1/2$ and $\delta_{3/2}(4) = 7/12$.

Proof: First, it is clear that a cycle of length 3 is a $3/2$ -clique and therefore we have $\sigma_{3/2}(3) = 3 \times 2/2 = 3$. By Lemma 2 and induction, we can show that there exists a $3/2$ -clique of size n with $n(n-1)/2$ arcs, i.e., $\sigma_{3/2}(n) \leq n(n-1)/2$, for any odd integer $n \geq 3$. Hence, by definition, $\delta_{3/2}(n) \leq 1/2$. In Lemma 1, we have shown that $\delta_{3/2}(n) \geq 1/2$, and therefore $\delta_{3/2}(n) = 1/2$ for

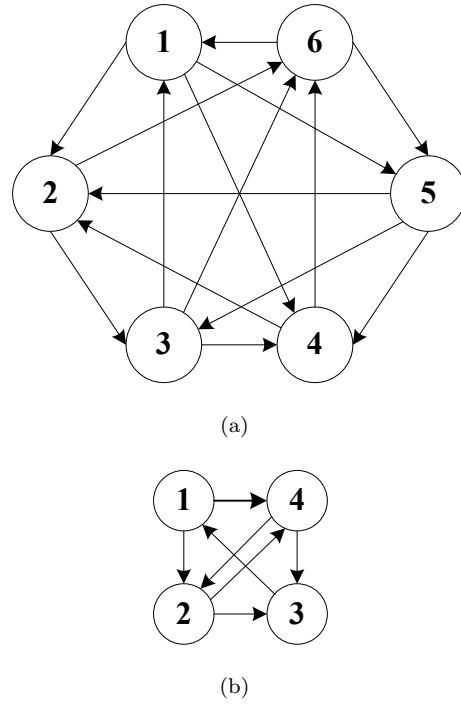


Figure 1: The minimum density $3/2$ -cliques of 6 and 4 nodes

odd integer $n \geq 3$. Similarly we can show that the result holds for any even integer $n \geq 6$ by giving a 6-nodes graph with $\binom{6}{2} = 15$ arcs as in Figure 1(a).

We have shown that the minimum density of a $3/2$ -clique for any integer $n \geq 3$ is $1/2$ except for $n = 4$. For $n = 4$, there is a $3/2$ -clique with 7 arcs as shown in Figure 1(b). Furthermore, by enumeration, we can show that there is no $3/2$ -clique with only 6 arcs. Therefore $\delta_{3/2}(4) = 7/12$. $\square$

It is somewhat surprised that the case of 4 nodes is special. But we may conclude that the density of a $3/2$ -clique is at least 0.5, and it is tight except for $n = 4$. The tightness of a lower bound means that the lower bound indeed happens but not only a loose estimate. This constant lower bound ensures a worst case quality of a $3/2$ -clique. We now check the minimum density of a 2-clique.

Lemma 4: $\delta_2(n) \leq 2/n$.

Proof: Consider a star-like network $G = (V, E)$, in which there is a center $x \in V$ and $E = \{(v, x), (x, v) | \forall v \neq x\}$. It is easy to verify that the distance between any two nodes

is at most 2, and it is a 2-clique. The number of arcs is $2(n-1)$, and the density is therefore $2/n$. $\square$

The above lemma gives a general result but $2(n-1)$ is not a tight bound for all n . The 3-node 2-clique with minimum density is a 3-cycle. It has 3 edges but not $2(3-1) = 4$ edges. In the next lemma, we show that any 2-clique has at least $2n-2$ arcs for $n > 3$.

Lemma 5: For any $n > 3$, $\sigma_2(n) \geq 2n-2$, and $\sigma_2(3) = 3$.

Proof: Let A be the adjacent matrix of an n -node 2-clique G , and the nodes are labeled from 1 to n . We denote $A[i, j] = 1$ if there is an arc from node i to node j . Let A^+ denote the adjacent matrix of the transitive extension of G . Since G is a 2-clique, $A^+[i, j] = 1 \forall i \neq j$. That is, we must have $A[i, j] = 1$ or $\exists k \notin \{i, j\}$, $A[i, k] = A[k, j] = 1$.

Let $B_i = \{j | A[i, j] = 1\}$ and $r_i = \sum_j A[i, j]$. For any i , we have

$$\sum_{j \in B_i} r_j \geq (n-1) - r_i$$

Let $m = \sigma_2(n)$ denote the number of arcs. Summing up for all i ,

$$\sum_i \sum_{j \in B_i} r_j \geq n(n-1) - \sum_i r_i = n(n-1) - m \quad (1)$$

Since $A[i, j]$ is either 1 or 0, the left-hand side can be rewritten as

$$\begin{aligned} & \sum_i \sum_{j \in B_i} r_j \\ &= \sum_i \sum_j (A[i, j] r_j) \\ &= \sum_j \left(\sum_i A[i, j] \right) r_j \\ &= \sum_j c_j r_j, \end{aligned}$$

in which c_j is the sum of j -th column. Since $r_j, c_j \geq 1$ (G must be connected) and $\sum_j r_j = \sum_j c_j = m$, the maximum of $\sum_j r_j c_j$ happens when $r_k = c_k = m - (n-1)$ for some k and $r_j = c_j = 1$ for any $j \neq k$. The maximum is $(m - (n-1))^2 + (n-1)$. Combined with Equation (1), we have

$$\begin{aligned} & (m - (n-1))^2 + (n-1) \geq n(n-1) - m \\ \Rightarrow & m \geq 2n-3 \end{aligned}$$

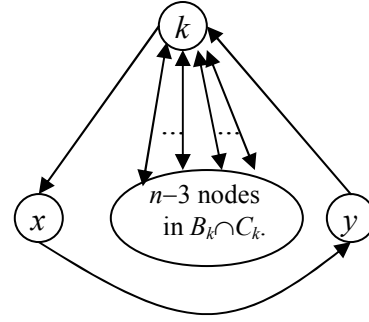


Figure 2: The case of $m = 2n - 3$

We have shown that $m \geq 2n-3$ for any $n \geq 3$. Now we show that $m \neq 2n-3$ for $n > 3$. Recall that the maximum of $\sum_j c_j r_j$ happens when $r_k = c_k = m - (n-1)$ for some k and $r_j = c_j = 1$ for any $j \neq k$. If $m = 2n-3$, then $r_k = c_k = n-2$. Let $C_k = \{j | A[j, k] = 1\}$. That is, C_k is the set of nodes having an out-going arc to node k and B_k is the set of nodes having an in-coming arc from k . Note that the sets B_k and C_k cannot be equal since all the nodes except k has exactly one in-coming arc and one out-going arc. Consequently there is exactly one node $x \in B_k - C_k$ and one node $y \in C_k - B_k$. It is clear that $A[x, y] = 1$ as in Figure 2.

If $n \geq 4$, there exists a node $z \in B_k \cap C_k$, and we can find that the distance $d(z, y) > 2$, which is a contradiction. So $m \geq 2n-2$ when $n \geq 4$. This completes the proof. $\square$

Theorem 6: For $n > 3$, $\delta_2(n) = 2/n$ and $\delta_2(3) = 1/2$.

Proof: By Lemmas 4 and 5. $\square$

3 Concluding remarks

In this paper, we define the 3/2-clique in a directed graph and show the minimum density is $1/2$. We also show the minimum density of a 2-clique is $2/n$. The result ensures the worst case cohesion of the groups found by means of k -cliques in social networks with unidirectional relationship.

To further test the approach of 3/2-clique, we performed tests on random data and real data. The real data is obtained by extracting 1000 sub-graphs from the friend network on Wretch Blog

[6], each with 1000 nodes. Each of the subgraph was obtained by snowball sampling starting from a random seed node. For these subgraphs, the average number of arcs is 5824. The average density of the subgraphs is therefore $5824/(1000 \times 999) = 0.00583$, rather sparse. For each subgraph, we computed a $3/2$ -clique by the randomized algorithm for maximum clique at the seed node. We recorded the size and the density, ignoring the cliques of size < 3 . We found that 35% of the maximum cliques have sizes ≥ 7 and 95.7% are of density ≥ 0.7 . Next, we performed tests on random data. We constructed 1000 random graphs with 1000 nodes and 5000 arcs. For each random graph, we found a $3/2$ -clique as large as possible at each of the nodes, and recorded the size and the density of the largest $3/2$ -clique. We repeated the experiments for random graphs with 1000 nodes but with 10 000, 20 000, and 40 000 arcs, respectively. We observed that these maximum $3/2$ -cliques are relatively small. Even in the graph with 40 000 arcs, there are few $3/2$ -cliques of size 6. Only 0.1% of the maximum $3/2$ -cliques have density ≥ 0.7 .

By these results, we may conclude that the Wretch friend network is well-structured and our method is appropriate.

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